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I. Executive Summary

The green building market is anticipated to be among the fastest growing industries worldwide[1]. Driven by citizen and building owner demand for sustainability and operational cost reduction, the green building market is, indeed, doubling every three years[2]. These trends are the main forces that are seeing a rise in the number of construction companies that are applying for green certifications for their projects in Europe-60% of projects by 2018[2].

To be green certified, a building project needs to fulfill a number of conditions, including safety and the use of sustainable materials. The strong market demand for green building solutions is in turn boosting the sustainable building material market, which is expected to reach USD 255 billion in 2020, growing at a remarkable CAGR of 12%[3].

MOGU is an innovation driven and environmentally conscious company that has developed a proprietary technology to produce natural-grown materials in response to the strong market demand for sustainable and high-performance materials.

Our MOGU technology uses specifically selected fungal strains-protected by Industrial Secret- for converting agricultural or industrial residual biomass into 100% natural, renewable and recyclable construction materials. MOGU material is both cost and performance competitive, in comparison with currently available solutions on the market (both bio and fossil based).

To design the most valuable applications for our materials, we have- with the support of our experienced Advisory Board and Mentors- performed a thorough assessment of the technology in different market case scenarios taking in consideration the following aspects: market segmentation and trends, entrance barriers, competitors, regulation, freedom-to-operate and IP landscape. This market analysis has shown the resilient flooring market to represent a great business opportunity for MOGU, and one that capitalizes on the competitive advantage of our technology. This market is indeed growing at a CAGR of 10.0%[4], driven by end user demand for simple, durable and sustainable housing solutions with innovative design qualities.

In order to tap into this opportunity, we have developed MOGU floor- an innovativeresilient tile for the flooring market- to TRL7. The development of this product has been performed in tight cooperation with stakeholders from the value chain: Scientific Partners (Utrecht University, Pavia University and Miconet), Technology/raw materials providers (CNC, Orineo, Resinalab and AEP Polymers), and Early adopters (ATI Projects, Incatro, ExT, Vaxana, Arup). The resulting outcome is a product that offers tailored low to medium density, is strong and resistant, shock-absorbing and is capable of outperforming the competing solutions already available on the market in terms of fire resistance, thermal insulation and noise absorption, in line with required standards and market benchmarks. Furthermore, the entire lifecycle of MOGU floor, including production, installation, maintenance and disposal, is dramatically less expensive than the competing solutions, and more environmentally friendly.

Our business vision is for MOGU to become one of the reference brands at international level with regard to innovation, sustainability and quality of living solutions. To realise this ambition, MOGU has elaborated an investor-ready business plan with the commercial target to sell more than 500.000 m^2 of material and to generate a cumulative turnover of 35 M€ over the next 5 years.

Our business plan has been evaluated by numerous acceleration programs and investors and has been awarded several prizes, including the Startup Initiative of Intesa San Paolo Bank as best Circular Economy business case, raising 150.000€ of private funding. The MOGU business case has also been presented as a highly promising initiative in various Bioeconomy and Circular Economy events, such as, IFIB, EFIB, Ellen McArtur Foundation, Ecomondo, World Forum for Responsible Economy, Klimahouse, among others.

Through this Phase 2 SME Instrument, we want to reduce the time-to-market for the commercialization of our products, so that we can capitalize on the competitive advantages offered by our brand both at a technical and strategic level.

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[1] Statista 2017

[2] World Green Building Trends 2016: Developing Markets Accelerate Global Green Growth SmartMarket Report - Dodge Data & Analytics

[3] Green Building Materials Market for Framing, Insulation, Roofing, Exterior Siding, Interior Finishing and Other Applications for Public facilities, Education, Commercial and Industrial, Healthcare, R&D Centers, Residential and Other End-users - Global Industry Perspective, Comprehensive Analysis, and Forecast, 2014 – 2020 - Zion Research

[4] Grand View Research, Inc

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1. Excellence

1.1 Challenge and solution

The identified customer pain point, business need, technological challenge and market opportunity

The demand for sustainable building solutions is currently the dominant force driving the building market at political, investor, constructor and customer level.

The need to limit the huge environmental impact generated by the building sector, has become a priority in almost all the political agendas worldwide. Several EU policies, such as the "Construction Products Regulation", the "Eco Design Directive", the "Green Public Procurement" and the "Resource efficiency opportunities in the building sector", along with sustainability standards for construction (such as CEN TC 350 and ISO/TC 59/SC17), are all aligned with the objective of reducing the environmental impact of buildings by improving their overall resource efficiency. The Circular Economy Action Plan of 2015 reiterates this objective and adds that, given the long lifetime of buildings, it is essential to encourage design improvements that will reduce their environmental impacts and increase the durability and recyclability of their components. In 2016, the EC commenced the debate for the forthcoming EU green building policy, with the aim of improving the sustainability, quality and value of the construction sector.

Property investors and construction companies are constantly seeking new solutions to achieve official green certifications that increase the value of the building and attract tenants and buyers. Several environmental certification schemes exist, a few of which are described in Table 1, all taking into consideration parameters such as resource efficiency, use of sustainable or recycled materials and indoor air quality. Investments in green certified buildings can produce measurable financial value, such as increased rental rates and asset value, reduced risk of depreciation, and higher tenant attraction and retention rates[5,6,7]. For building owners, a green certification increases the Return-on-Investment (RoI) and the value of their building by up to 19.2%[8]. A world-wide study covering 69 countries reported that 40% of construction companies are driven first by the market demand for green buildings. In EU, construction companies are going to apply certified green solutions in more than 60% of their projects by 2018[9].

Table 1. Main green building certification schemes

Scheme	Description
LEED	Probably the best-known of all green building certifications, LEED (Leadership in Energy and Environmental Design) is a system established by the U.S. Green Building Council to measure and rate the environmental performance of buildings and building interior. The LEED system verifies the sustainability levels of whole built environments by awarding points for excellence in areas such as energy usage, water efficiency, use of recycled materials and indoor air quality, which are totaled to reflect different levels of certification.
BREEAM	The Building Research Establishment's Environmental Assessment Method (BREEAM) was the first comprehensive green building rating system in the U.K. In order to achieve a BREEAM rating, a building must undergo an assessment that measures a number of factors including energy and water use, the health and well-being of inhabitants, pollution, transportation issues, materials, waste and ecology and management processes.
HQE	The HQE label, founded in France in 2002, aims to improve environmental quality of new and existing buildings. The certification aims at making sure the infrastructure is contributing positively to the respect of the environment, the social and economic life and the planning of a territory. This certification enables listing all the questions of project owners committed to taking into account sustainable development in their infrastructure projects Identifying territory stakes and projects, leading an in-depth consultation.
DGNB	The DGNB System has been created in Germany in 2008. Quality is assessed comprehensively over the entire life cycle of the building. Due to its flexibility it can be tailored precisely to various uses of a building and even to meet country-specific requirements. The fulfilment of up to 50 sustainability criteria from the quality sections ecology, economy, socio-cultural aspects, technology, process work flows and site are certified.

[5] Dow Jones Sustainability Index, SAM, 2010
[6] Ethical Stock Indexes: Does Sustainability Pay Off? C. Consolandia, P. Nascenzib, A. Jaiswal-Dale, Italy, US, 2008
[7] Doing green by doing good? An analysis of the financial performance of Green Office Buildings in the USA, RICS research, 2009
[8] McGraw Hill Construction (2010). Green Outlook 2011: Green Trends Driving Growth.
[9] World Green Building Trends 2016: Developing Markets Accelerate Global Green Growth SmartMarket Report - Dodge Data & Analytics

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Last but not least, building operators and house tenants are increasingly looking for green buildings, motivated by economic, environmental and social reasons. According to the "World green building trends 2016" Report[9], the first reason for choosing a green building is the possibility of reducing conduction and maintenance costs. Indeed, the benefits of green solutions for building tenants, include decreased operating costs up to 13.6%[7]. In addition to the economical factor, reduced resources consumption, encouraging sustainable business and providing healthier indoor environments are the most important environmental and social reasons for choosing a building green.

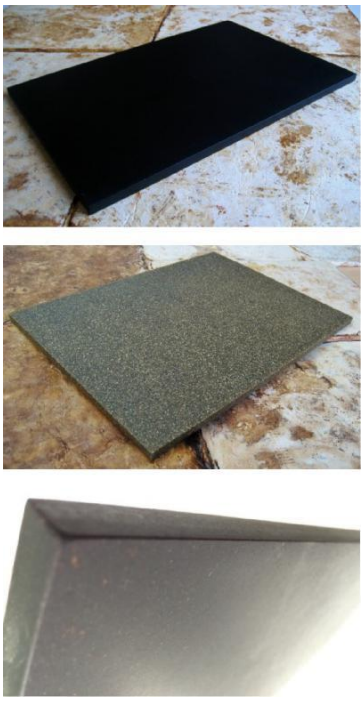

Fig. 1: MOGU floor prototypes

Buildings do not only affect the natural ecosystem, but also human wellbeing- both physical and mental. End users are now looking for more functional, healthy and gratifying living experiences, both in commercial and residential buildings. Green solutions shouldalso be durable and carefree, easy to install, clean and maintain and should provide the end user with novel and seamless functionalities, in an intuitive and fluid way. Finally, they should be stylish, offering an appealing design and greater degree of customization in terms of colours and materials in order to creating pleasant and cozy spaces.

In conclusion, today we are seeing multiple factors converge, escalating the need for sustainable and functional building solutions, resulting in the green building market being one of the fastest growing and interesting marketsforthe coming years.

Our Innovation

Increasing demand for functional and easy-to-live-in building solutions is the main driver of the impressive growth of the resilient flooring market. Indeed, resilient flooring is the fastest growing segment of the flooring market, with a CAGR of 10% in comparison with the 4% of the overall flooring market[10].

Unlike conventional hard flooring, resilient flooring is made of materials that offer a certain level of elasticity, giving the flooring a degree of flexibility, known as resilience. Resilient flooring is extremely easy to maintain and durable, water resistant and inexpensive, and for these reasons it is so appreciated by end users. Recent technological innovations allow a natural mimicking layer that reproduces natural textures, such for example wood, to be printed on the surface of certain types of resilient flooring. This feature is resulting in a great success of this market segment, which is contributing to the growth of the resilient flooring market.

To meet the market demand for sustainable, functional and beautiful living solutions, we have developed MOGU floor, a natural-grown and high-performance resilient flooring. Our flooring is 100% natural, renewable and recyclable (see Fig.1). It is a green building product designed to deliver great value to end users and consumers, satisfying their demands for sustainability, safety, ease of use, performance and design (see Tab 2).

<u>Sustainability</u>: MOGU floor is a 100% biobased and natural product. Its manufacturing is performed under mild conditions, with a very low carbon footprint of the entire production.

In addition to being 100% renewable, MOGU floor also meets the principles of the Circular Economy, reducing waste generation at a systemic level. We produce MOGU floor starting from byproducts of industrial processes, such as textile fibers, co-products of wire manufacturing, or leftovers such as sawdust, straw, wood, fiber etc. The use of byproducts as raw materials brings two main advantages: reduction of the feedstock cost and reduction of the environmental impact and costs related with the disposal of the former byproduct. MOGU is collaborating with the cotton wire manufacturing company CFT Masserini (Bergamo - Italy) that produces 3 ton/day of waste fibers (sufficient for producing around 1000 m^2 of MOGU floor, which is more than sufficient to cover our commercial targets). These waste fibers represent a very suitable feedstock. This material- utilized in the past decades for the production of hearing sticks and adsorbent

[10] Resilient Flooring Market Analysis by Product (Luxury Vinyl Tiles, Vinyl Composite Tile, Vinyl Sheets, Fiberglass Sheets, Linoleum), By Application (Residential, Commercial, Industrial) And Segment Forecasts To 2022-Grand View Research, Inc. 2015

product- has become a waste during the last decade with Associated**negative** with**consequences**documentRef.**at**Ares(2018)3790472**economicand** - 17/07/2018 **environmental level. Moreover, MOGU floor has a sustainable end-of-life, being recyclable or compostable in industrial composting plants (according to standard EN 13432: 2000).**

Table 2. MOGU floor characteristics

Characteristic	Evaluation	Standard
Biomass content	Up to 100%	ASTM D6866
Panel size	600 x 600 mm 500 x 500 mm 300 x 300 mm	
Thickness	8 - 10 mm	
Fire resistance	C, S1	EN 13501-1
Thermal conductivity	0 - 100	EN 12664
Acoustic insulation	30 db	
Slip behaviour	Safe surface	EN 13893
Abrasion	No damage	EN 425

Taking in consideration the entire lifecycle of MOGU floor, it is probably one of the most sustainable flooring on the market. As a comparison, the linoleum flooring commercialized by Forbo, one of the market leaders in the sector, is claimed as "the most sustainable choice available". Forbo's product is made from 97% natural raw materials, 70% of which are renewable, along with a 43% recycled content[11]. MOGU floor outperforms "the most sustainable choice" currently available on the market in that we provide a product that is 100% natural, 100% renewable and contains more than 80% of recycled matter.

Safety: The fast growth of the resilient flooring market is led by the increase in the sales of Luxury Vinyl Flooring (LVF), a type of flooring made of different types of materials all containing polyvinyl chloride (PVC). Despite its popularity, LVF has been reported to be potentially harmful to people and the environment as a result if its manufacture, through to its installation and use, and until its disposal at the end of its lifecycle. During its production, vinyl flooring generates large quantities of persistent toxic organochlorines such as dioxins and phthalates, which are highly persistent, bioaccumulative and toxic to humans and ecosystems. Vinyl flooring has been reported as potentially releasing these hazardous chemicals in the indoor environment over the years[12,13]. As interior spaces have limited ventilation, even small amounts of emissions can accumulate over time and create unsafe living conditions. For this reason, Indoor Air Quality (IAQ) is the main safety parameter to take in account for any flooring product, and in particular the emissions of Volatile Organic Compounds (VOCs). VOC emission can lead to: eye, nose, and throat irritation; headaches, loss of coordination, nausea; and damage to the liver, kidney, and nervous system.

Many countries have established regulations for limiting the concentration of VOCs, for example the AgBB in Germany or the ANSES in France or the CDPH in California, and in Europe, since 2011, the EU-LCI working group with the aim of recommending EU-wide harmonised health-based values for the assessment of product emissions.

Moreover, the main green building certifications have specific limits and protocols for measuring VOC emissions. BREEAM prescribes the limits for VOC emissions for flooring (compliance with EN14041:2004 in relation to declaration of Formaldehyde emissions) and adhesives (verification that carcinogenic or sensitising volatile substances are absent with evidence being provided by testing to EN13999). Similarly, in 2006 the US Green Building Council agreed to consider resilient flooring emissions as one of the parameters for a LEED credit (IEQc4.3).MOGU floor, unlike vinyl flooring, is 100% biobased, does not contain organochlorines and is bio- and chemically safe.

Ease of use:Consumer demand for functional and innovative solutions is the second force driving the building market.It is one of the main reasons behind the success of resilient flooring, which, in comparison to conventional flooring, is also more durable, shock and noise absorbent, carefree and thermal insulating.

However, resilient flooring is affected by demanding installation and poor aesthetics. Typical resilient flooring, such as linoleum or PVC-based (vinyl hereafter), is installed as unibody thin flexible sheets (2.0 - 2.5 mm in thickness) directly on the concrete slab (the flat piece of concrete that serves as a walking surface in

[11] https://www.forbo.com/flooring/en-gl/products/linoleum/cmlack

[12] Xu Y, Cohen Hubal EA, Little JC. Predicting Residential Exposure to Phthalate Plasticizer Emitted from Vinyl Flooring: Sensitivity, Uncertainty, and Implications for Biomonitoring. Environmental Health Perspectives. 2010;118(2):253-258.

[13] Mendell M, J. Indoor residential chemical emissions as risk factors for respiratory and allergic effects in children: a review.indoor air international journal of indoor environment and health 2007;17(4):259–277

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buildings). To ensure proper installation, the concrete slab must be perfectly flat, otherwise the thin and flexible flooring material will follow any imperfection of the ground, resulting in an irregular surface.

In addition, linoleum and vinyl flooring are highly sensitive to humidity. When humidity increases, due to climate conditions, the adhesive holding the floor in place weakens causing brittle cracks. This combines with curling sheet edges to create a floor that loosens in some spots. To limit this drawback, the flooring industry has agreed on a standard criterion for the acceptability of a concrete slab to receive installation of resilient flooring materials. The acceptability is based on the results of a Moisture Vapor Emissions Rate (MVER) test performed on the slab surface in accordance with the standard method ASTM F 1869, "Standard Test Method for Measuring Moisture Vapor Emission Rate of Concrete Subfloor Using Anhydrous Calcium Chloride". The installation of resilient flooring cannot begin, and warranties on flooring materials will not be honored, if MVER results are greater than 1500 gr/100 m^2/24 hours. This limit of acceptability is also adopted in ASTM F 710-11, "Standard Practice for Preparing Concrete Floors to Receive Resilient Flooring".

Further limitations of conventional resilient floorings are characterized by high maintenance costs and low reparability. In case of damages, caused for example by the fall of an object, it is not possible to fix the affected area without removing and reinstalling the entire flooring.

MOGU floor has been designed and developed to resolve all these limitations and to deliver an easier and more enjoyable user experience to the customer. MOGU floor is not a large thin flexible film, but a solid thick tile. This high-quality design allows the tiles to be installed in a modular setup, regardless of the humidity conditions and without the need to prepare a perfectly flat concrete slab. The thickness and relative rigidity of the tile compensate any imperfections of the concrete slab,as well as any potential weak points of the adhesive used for the installation. Moreover, MOGU floor is easy to clean and waterproof and maintenance requirements are minimal.Inthe event of damage, the modular design allows the affected tile to simply be replaced on the fly, as has been confirmed by Orineo (resilient floor provider) during preliminary tests performed to date.

To showcase how MOGU floor simplifies and improves the user experience, the installation of a linoleum flooring on a 60^{2}m surface would cost 3000 - 4200 € and take 6 hours (including concrete slab preparation, MVER testing and product application).The same installation of MOGU floor would cost 2000 - 2400 € and take only 2 hours.

Design:Conventional resilient flooring is generally not used in living environments due to its unappealing appearance and low personalization possibilities. In recent years, new products such as luxury vinyl flooring have taken advantage of photorealistic printing solutions to improve their aesthetic by simulating natural textures and in this way achieving impressive commercial success.

MOGU floor takes the design possibilities of resilient flooring to a new level, thanks to its natural look and feel. As opposes to a printed layer, the different types of raw materials used in the manufacture of MOGU floor creates different textures and patterns that can be actually touched and felt with the hand. Furthermore, during the finishing step of the manufacturing process, additional materials are added, in the form of particles, with resin like coconut shells or coffee to create an even more complex texture and natural feel that can be enjoyed at every step.

The combination of different raw materials, finishing particles, colors and sizes, affords MOGU floor with tremendous personalization possibilities, capable of satisfy the needs of even the most refined end user (such as hemp, miscanthus, corn, sunflower, soy straw), with different finishing (bio based resins) and colors (using different biomasses as fillers, to obtain black, red, green, brown, and yellow) and size (350x350, 500x500, 600x600 mm).

Price: In terms of price, MOGU floor is cost competitive with the other solutions on the market, and taking in consideration also the costs of installation and maintenance, it is particularly affordable, as described in Section 2. Just to give an example, including installation, luxury vinyl flooring costs 80-110 €/m^2, which is more expensive than MOGU floor, which costs 80-90 €/m^2. All in all, MOGU floor delivers outstanding value for money to the end user and a unique user experience, in terms of ease and living experience.

Why MOGU floor represents a breakthrough beyond the market state of the art, and how it brings benefits to the environment and society

Our proprietary technology has the potential to completely redesign the current value chains of the building market, allowing the use of a wide range of raw materials, currently considered unsuitable for performance

demanding applications. For the first time it is possible to convert left overs and byproducts into high-performance material and in this way increase the environmental and economical sustainability of the flooring market, as well as of the value chain from which the residues originate.

Fig. 2: One of the press in MOGU

Currently only a few building materials contain at least a fraction of recycled materials that are used for low value applications such as wall cavity fillers for insulation. Indeed, the use of recycled materials tends to decrease the performance parameters, such as the compressive strength and the modulus of elasticity, of the final product. For this reason, the recycled materials on the market are used in low value applications,such as fillers.

MOGU technology surpasses the current limits of recycling, allowing the raw material characteristics and final performance to be upgraded. This is possible thanks to the biotechnological process, driven by specifically selected fungal strains capable of converting the raw material into a high performance composite material. The fungal metabolism changes the physical properties of the raw material and creates a dense network of roots, called mycelia, within the feedstock. Under controlled conditions the fungi grow until transforming the heterogeneous matrix of the feedstock into a solid material with low or medium density, which is also strong, resistant and shock-absorbing.

In addition to the biotechnological innovations, we have also developed engineering solutions to improve the final performance of MOGU floor. Following the fungi growth phase, the intermediate product undergoes a series of drying and pressing steps, and then is finished by adding an up to 100% biobased resin before the final surface treatment to make it water resistant, easy to clean and more durable. To implement these manufacturing steps, we have had to modify and adapt a commercially available industrial press to our specific needs (the customization work has been performed in a dedicated research program with the collaboration of Utrecht University and TU Delft- refer to Fig.2) and we had to test several combinations of resins and coatings to achieve the desired performance.

All in all, MOGU floor is an unprecedented engineered flooring that offers numerous benefits that competitors simply cannot match in terms of sustainability, performance (see table 8), style and design.

Indeed, the building sector is responsible for about one third of all waste produced in Europe and is associated with environmental pressures that arise at different stages of a building's lifecycle[14]. Despite the environmental impact of this sector and the efforts of the EC to support the transition towards a biobased and circular economy, the availability on the market of green materials for construction is still low and even lower in the case of solutions recycled or produced starting from byproducts.

At MOGU, we have the ambition to develop a new circular feedstock supply chain in the building sector and deliver an innovative and high-performance material of great design. In this context, MOGU floor is a unique example on the market in terms of sustainability, performance and innovation. It is made from natural-grown material, using byproducts of other sectors, and transformed to reach the highest quality standards of the sector. The result is unique also in terms of performance of the final products. MOGU floor outperforms the competitors in a multitude of aspects, while coming in at a price that is up to 30% less than competing products.

In addition to commercializing in Europe, our ambition is also to bring MOGU floor to the Asian and American market, by licensing our technology to local players in these regions. The licensing would be time and market limited so as to not prevent the possibility of us opening new production plants in these regions and directly commercialize MOGU floor, or future developments, there in the future.

1.2 Approach

What is unique in our approach and themilestones that have led to the current development stage and its unique selling points

Our MOGU proprietary technology is the result of 3 years of developments in the fields of biotechnology and materials science, for growing functional composite materials directly from residual substrates generated

[14] Transition to Sustainable BuildingsStrategies and Opportunities to 2050- International Energy Agency- 2013

during agricultural or industrial processes. The resulting technology holds tremendous potential to radically change the current sustainability standards of a wide range of industries and to improve the profitability of the related value chains. Thanks to our 24 months workplan (described in Section 3.2), we will bring the Circular Economy concept to the construction sector, as well as to other sectors in the future as we transfer our technology to additional sectors.

The MOGU technology can be described in three major phases, summarized in Fig.3:

Fig. 3: MOGU floor life-cycle

1) Growing Phase: This is the stage of the manufacturing process in which the material is alive and growing. The feedstock is firstly sterilized and then the fungal inoculum takes place in aseptic conditions. Both the type of feedstock and the fungal strain will deeply affect the final product characteristics. The inoculated feedstock is placed in hermetic bags and the conditions adjusted in terms of humidity, pH, and gas exchange. After one week the substrate is ready to be removed from the bag, grinded and then placed into a mold where it will growth for two more weeks, reaching the physical status of a wet bio-panel.

2) Pressing and drying phase: Once the wet bio-panel reaches the optimal density, it has to be dried and shaped into the final product format. This phase is performed using a suitable press, usually heating the plates accordingly with the moisture content of the wet bio-panel (about 50%). The hot-pressing technique obtains a tile that totally dry, and biologically stable (the mycelium is deactivated by the heat).

3) Finishing Phase: During the last step of the process, the semi-finished product undergoes finishing in order to arrive at the final distinctive look and feel. The biobased resin is applied together with natural elements such as fragments of roasted coffee bean or coconut shell. Then the tile is painted, coated with a protective layer and sanded.

Why now? Current Scenario and Future Trends of the Flooring Market

The green building market is doubling every three years and it is expected to reach USD 15.88 trillion by 2024 due to the increased market demand for green solutions and the national push toward the adoption of green building practices[15]. At the same time the flooring market is also growing, mainly driven by the end user demands for functional, carefree and attractive products.

In recent years, end users demand is deviating from traditional hard flooring in favor of the innovative solutions introduced in the resilient flooring segment. The global resilient flooring market size is indeed projected to reach USD 92.47 Billion by 2022, growing at a CAGR of over 10.0% from 2015 to 2022[14], mainly driven by the Luxury Vinyl Flooring (LVF). The latest products launched in the LVF segment are dominating flooring sales by closely mimicking the look of natural materials, such as wood or stone, thanks to a surface layer that has a very realistic photo-created image, combined with textures that resemble the real materials. The willingness to pay for such products place these solutions in the high price or Luxury tier of the resilient flooring market.

The co-occurrence of the impressive growth of both green buildings and luxury resilient flooring markets, opens a huge market opportunity. MOGU floor is made with the most sustainable building material currently available in the construction sector and is able to outperform the competing products in the luxury flooring market in terms of performance, sustainability, price, design and ease of living (as reported in table 8). In this highly favourable market scenario, an additional factor that will foster the market uptake of MOGU floor are the growing efforts of the major governments to support the transition towards the Bioeconomy with regulations, such as the "Construction Products Regulation", the "Eco Design Directive", the "Green Public Procurement" and the "Resource efficiency opportunities in the building sector", along with sustainability standards for construction (such as CEN TC 350 and ISO/TC 59/SC17), and several national fiscal subsidies. To take advantage of this window of opportunity we are now applying to the SMEI Phase 2.

[15] The U.S. Market for Green Building Materials - BBC Research 2015

At the current development status, MOGU floor is at TRL7, as explained in table 3. The more suitable fungal strains have been selected according to the types of raw material used, the bio-panel growth and characteristics have been characterized and the manufacturing process to convert the bio-panel into the final product has been developed. As a result MOGU floor prototypes have been developed in our pilot facility (see section 4-5), and also based on the feedback of stakeholders from the green building sector such as Arup, ExT, Incatro, AtiProject and Vaxana. The prototypes have also undergone a preliminary verification for compliance with the CE mark needed for commercializing a building material in Europe, according with the methods in table 4 and as shown in Fig 4.

Table 3: MOGU floor Technology and Manufacturing Readiness Level

Element	Current TRL	TRL after SMEi	Current MRL	MRL after SMEi
Selection of fungal strains	7/8	9	8	9
Bio-brick growth	7	9	7	9/10
Production of the semi-finished product	6/7	9	6/7	9/10

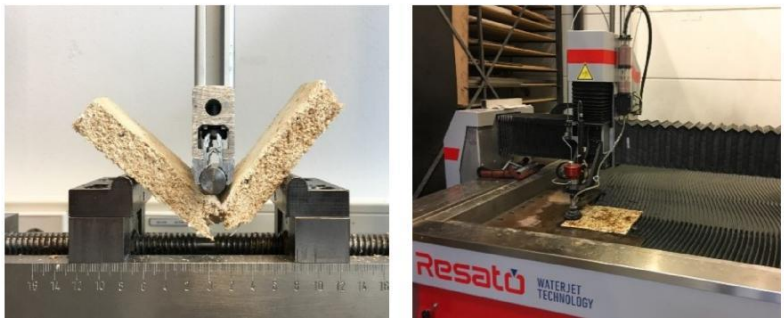

Fig. 4: MOGU material testing and cutting

Regarding the Manufacturing Readiness Level, MOGU floor is currently at MRL7. At our pilot facility we are indeed capable of manufacturing 50 m^2 per day of MOGU floor in a production representative environment. Our commercial strategy aims at high volumes of sales and the manufacturing readiness is therefore crucial for the business viability. From this point of view, the raw material specifications have been approved and thematerials are available to meet the planned production schedule. We have secured the procurement of raw material from CFT Masserini Spa for the supply of textile fibers, resulting from the production of cotton wires, which are an ideal feedstock for the production of MOGU floor.

Table 4: Standard methods used for preliminary testing of MOGU floor

Fire classification of construction products and building elements - Part 1: Classification using data from reaction to fire	UNI EN 13501-1:2009
Water quality - Gas chromatographic determination of chlorophenols in water	EN 12673:1998
European Regulations for Formaldehyde	EN 717-1
Building materials and products - Hygrothermal properties - Tabulated design values and procedures for determining declared and design thermal values	UNI EN ISO 10456:2008
Resilient, laminate and textile floor coverings - Measurement of dynamic coefficient of friction on dry floor surfaces	UNI EN 13893
Resilient and laminate floor coverings - Assessment of static electrical propensity	UNI EN 1815:2016

We have also reached a good reproducibility and yield of the process and therefore we are expecting a time-to-market of 18 months from the beginning of the SME instrument.MOGU floor prototypes have already been demonstrated in several installations and promotional events, such as the world famous Milan Design Week 2017.Currently MOGU has a collaboration agreement with the University of Pavia and Utrecht University for the identification of new fungal strains, which includes strict exclusivity and confidentiality clauses. For the products finishing, MOGU has established suitable partnerships with technology providers in the bio-resins and sustainable flooring sector, such as Orineo, AEP Polymers and Resinalab. Incatro, specialized in acoustic design, has encouraged the application of the MOGU material for contexts where acoustic performance is required (public buildings, hospitality). In addition, our internal MOGU development program is in the process of testing new applications, such as acoustic insulating panels (currently at TRL5) or new products for the automotive sector (currently at TRL4), which will fuel long-term future business lines.

To finalize the development of MOGU floor, and to launch our first collection on the market, we will implement a market oriented workplan aimed at implementing our commercialization strategy and preventing the related market risks. To this end, we have elaborated a SWOT analysis, summarized in Table 5, to identify technical and commercial risks and opportunities related to the successful market introduction of MOGU floor.

Table 5: MOGU floor SWOT analysis

Strengths	Weakness
Innovation: MOGU technology is a unique and unparalleled conversion method of residues into advanced materials. Sustainability: MOGU floor is probably the most sustainable material on the market and the only biobased circular product in the flooring market. Business Model: Our business innovation project will redesign the flooring market value chain in a truly circular approach. Performances: MOGU floor offers in a single product functionality, ease of use, and performance that competitors cannot match. Design: MOGU floor has a natural texture and pattern that provides a unique look and feel on the market. Safety: MOGU floor is free of any potentially hazardous chemicals and contributes to healthy IAQ. Price Positioning: MOGU floor is a top notch product sold at a price typical of lower market segments. Network: MOGU team has been very active in the building market and has developed a very relevant network with key stakeholders.	Production Capacity: whileMOGU floor is at TRL7, our production capacity is currently limited. For this reason, to match our commercial target we will upscale our production plant. Investment need: the development stage of MOGU floor is in the Valley of Death. For this reason we developed a fundraising strategy in which the SMEI is pivotal. Positioning of competitors: MOGU is a new player in the flooring market. To bridge the gap with competitors we are focusing on influential early adopters to drive the market uptake of our product. Small team: While our current team is highly prepared and motivated, we need to grow the MOGU team to deploy our commercial strategy. To this end, we have elaborated a recruitmentprogramme.
Opportunities	**Threats**
Market demand: the green building and resilient flooring markets are both growing at an impressive rate. MOGU floor is able to satisfy the market demands of these two sectors and take advantage of a great market opportunity. Policy: the reduction of the environmental impact of the building sector is a priority at National and International level. MOGU floor will benefit from new regulations in favor of more sustainable buildings. Certifications: MOGU floor can be granted with different green schemes in place in many countries. A strategic selection of green certification will open new markets to MOGU floor. IP: the FtO study performed by our attorney has highlighted the potential patentability of MOGU technology. This could open new IP strategies to license MOGU technology in Asianand American markets. Industry 4.0: we will implement digitisation and data-driven Industry 4.0 approaches in our production plant. This will allow us to run a highly efficient process and optimize it according to the different raw materials.	Price competition: some direct competitors may respond to MOGU floor success by reducing the price of their products. MOGU floor- even if cheaper than the current competitors- will not compete on price but on quality and functionality. Delay in certifications: Green certifications are capital in our commercial strategy. Any delay in their achievement may affect our business plan. For this reason we have separated the mandatory CE mark, from the other optional certifications. In addition, we will perform a preparatory study to evaluate which certification to apply for, to avoid themost risky labels. New Technologies: Resilient flooring is a very active market in which new solutions are continually being developed year on year. For this reason,TechWatch is a continuous activity to ensure we are informed. Volatility of Raw materials: The feedstock for our process can vary in price and availability. For this reason we validated our technology with different residues and we have already signed long term supply contracts.

We have performed a deep self-assessment of our innovation, at a technological and business level, and we have elaborated a business strategy for the successful commercialization of MOGU floor. The approach of our work plan aims at consolidating our competitive advantages and filling the current gaps of MOGU floor.

On the one hand, and in order to deliver the greatest possible value to the end user, we will select the optimal mix of functionalities and designs for MOGU floor, in addition to the most valuable green certifications to apply for. In parallel, we will sign collaboration agreements for the early adoption of MOGU floor with relevant stakeholders of the building sector. Finally, to protect our innovation and to open new commercial possibilities we will select and apply for the most suitable IPR protection instruments.

On the other hand, we intend to upscale our production plant implementing digitisation and data-driven approaches to allow us to be as efficient and cost effective as possible in meeting our production targets. At

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the same time, we will apply for the most suitable green certifications in parallel to reduce the risks of delays. Finally to support the growth and scaling of our company, we have elaborated a recruitment programme to help us to attract and retain the necessary talent for having a high-end, comprehensive experienced workforce.

A key value add in our approach, is that we will work in co-creation approach with key stakeholders, such as AEP Polymers, Orineo, and Resinalab, which is also member of the Italian Green Building Council, and is supporting companies and customers regarding the green certifications (LEED, BREEAM, etc.). Letters of Support have been annexed to the end of this bid.

Specific Objectives, the expected outcomes of this Phase-2 project and the related success criteria

The SME Instrument (SMEI) is an important element of our fundraising strategy. It represents a great opportunity for our company to bridge the remaining gap that is now dividing MOGU floor from the market, while keeping the ownership of our technology and retaining our equity.

To close this final gap, our aim is to achieve the following specific objectives over the course of a 24 months workplan from this Phase 2 SMEI (described in Section 3.2):

1. Define the specifications of the first MOGU floor collection.

Different prototypes of MOGU floor have already been realized and tested. To maximize our value proposition, and meet the flooring market demand, we have planned a co-design approach with stakeholders, customers and end-users to select the best set of specifications for MOGU floor. Specifications will include design parameters, both technical and aesthetic, and also the environmental certifications to apply for. This latter aspect is pivotal in our commercialization and internationalization strategy and for this reason a thorough review and selection of the currently available green and quality certifications will be carried out.

Performance indicator: At M4 the complete set of products specifications and a product collection of more than 10 customizations is defined (MS1). At M4 the list of green certifications to apply for will be defined (D1.1).

2. Optimize and upscale the manufacturing process of MOGU floor

The production process will be scaled up and optimized for efficiently meeting the final product specifications. The improvements will involve the raw material pretreatment, the growing conditions of the fungi on the raw material, the manufacturing process of the semi-finished products and its final treatment. All the parameters affecting these stages will be assessed in the current pilot plant at MOGU's facility in order to obtain meaningful results for the upscaled production.

The current maximum productive capacity of our facility is 50 m^2 of tiles per day. To increase our production capacity and reduce the amount of losses along the process, we will develop a high-performance production system. To be able meet our commercial target of 0.05% share of the European market (equivalent to 180.000 m^2), production capacity will be increased to over 600 m^2 per day. An advanced spectroscopy-based monitoring system will be developed and installed to control the Critical Process Parameters (CPPs) along the production line. In keeping with Industry 4.0 principles, the data acquired by the process monitoring system will be processed using embedded Artificial Intelligence (AI) algorithms and a predictive model will be developed. The model will provide information to adapt the process conditions to fluctuations and, thanks to the implementation of Machine Learning techniques, will be capable of adapting to future developments of the process including new raw materials or fungal strains.

Performance indicator: By M6 the optimized protocols for the MOGU products will be delivered (D2.1). By M18 the first commercial production of the MOGU material will reach the market (MS5).

3. Increase the value of the MOGU products by obtaining quality and environmental certifications.

Part of the value of MOGU floor lies in the green certifications it can provide to our client (especially for building companies, which are one of our target customers). Green certifications are instrumental to our business strategy and our expansion into international markets. As such, obtaining the selected certifications is an important objective for accelerating the commercialization of MOGU floor. During the SMEI project, we will obtain the CE marking and a number of strategic green labels to increase the value of MOGU floor.

Performance indicator: By M13 and M18 the CE mark and the additional green certifications will be achieved (MS7 and MS8 respectively).

4. Reinforce MOGU's Intellectual Property Rights (IPRs) and Brand Identity.

Our freedom-to-operate analysis has confirmed that the MOGU products are not infringing on any existing third party IPRs. Moreover, this study revealed that parts of the MOGU manufacturing process are indeed novel, such as the blending with resins. Non-disclosure agreements and material transfer agreements have already been signed with the current suppliers and involved stakeholders. These preparatory actions will be used for filing the patent applications that will allow the licensing of our technology in Asia and America, as envisaged in our commercialization strategy. In addition, and in line with our vision for ensuring that MOGU becomes a reference brand in the market, we will register the MOGU trademark to protect the visual identity of the MOGU brand and to help end users in identifying MOGU floor.

Performance indicator: By M18 IPR assessment is performed and any IPR applications filed (D5.4).

5. Engage influential stakeholders as early adopters of MOGU floor.

One of the key actions of our commercialization strategy is to engage influential early adopters in the building sector, to acquire reputation and visibility and quickly gain market share. Design and architect studios, real estate consultants, etc., are key stakeholders involved in designing the most prestigious and valuable building projects. Such projects are used by the market to define new trends and reference points. For this reason, engaging such stakeholders as early adopters of MOGU floor will represent a great boost for our market uptake. We have already signed some agreements with influential stakeholders such as ARUP, ATI projects and ExT (see letters of support provided in the Annex at the end of this bid), and additional ones will be signed during the SMEI project.

Performance indicator: At M18 agreements with early adopters will be signed (MS12).

6. Review our business strategy and launch MOGU floor.

We have elaborated and improved our business plan and strategy thanks to our participation in Acceleration Programmes that we won places on, and with the support of our mentors. During this project, we will review our Business Plan, and updated it on the basis of the results and the developments achieved during our workplan. Early adopters, distribution channels and initial markets for starting our internationalization, are among some the variables that will be updated. Marketing actions will be implemented accordingly, to maximize the impact of the SMEI project and to prepare for market launch of MOGU floor. We will plan our participation at the most relevant events of the building sector for high visibility and engagement with potential clients, and we will also submit publications in international journals, as well as design and implement impactful social media campaigns,in addition to creating promotion materials to raise the awareness about MOGU floor among our target audiences.

Performance indicator: By M3 revamped website (D5.1), by M6 dissemination/communication plan (D5.2), by M9 market review (D5.3) and by M24 updated/investor-ready business plan (D5.5).

2. Impact

2.1 Entering the market

Introduction to the market

In terms of market size, Europe accounts for over 35% of the overall resilient flooring shipments, which can be attributed to the presence of several key manufacturers in the region coupled with a high level of awareness. The flooring market in Europe is worth approximately 50 B€, out of a global market of 300 B€.

The global resilient flooring market is dominated by a few key market players: Tarkett (France, 2.4 Billion €), Forbo (1.2 Switzerland, Billion Swiss Francs), Beaulieu, Mohawk, Gerflor, DLW Flooring, Nora Systems, James Halstead, Amtico. The main customers of these players are construction companies.

In EU around 820,000 enterprises are active in building construction, most of which are micro and small enterprises. Among the largest enterprises, the top ones are Vinci (France, €38.5B revenue), ACS (Spain, €34.8B), Bouygues (France, €33.1B), Skanska (Sweden, €15.7B), Eiffage (France, €14B) and Strabag (Austria, €12.4B). The construction sector contributes around 9% of EU GDP, with more than 3 million companies and a total direct workforce of almost 13 million people. Since 2015, the construction industry has been recovering and Euroconstruct foresees a record growth rate of 3% for 2016 (as opposed to 2.4%in 2015) and 2.7% for 2017. Among the Member States for which data are available, the highest growth in construction was in Slovakia (+18.8%), Spain (+13.0%) and Sweden (+11.3%).

To have a complete picture of the green building market, however, it is important to consider that it not only affects the immediate customers, but also impacts on a broad range of other stakeholders. To simplify this complex scenario, four stakeholder groups have been clustered:

Group 1 includes end users, distributors and suppliers, who have greatest impact in determining the success

of a product or project. The feedback of qualified suppliers can A ciated positively with influence document Ref the. Ares(2018)3790472 perceptionofa - 17/07/2018 products and collaborations with distributors will increase the credibility of a firm's actions. This specific stakeholder group has been pivotal in MOGU floor development.Group 2 includes the press and media, environmental organizations, and the local population. These stakeholders can affect the project reputation by "mobilizing public opinion for or against the project performance". Particular efforts will be spent to engage such stakeholders for promoting MOGU floor and increase the market demand.Group 3are the "providers of critical inputs", which include clients, financial organizations, labor unions and professional associations. It includes also insurers who can grant better insurance conditions for sustainable buildings, certifiers who assess the sustainability performance of buildings, and researchers who can study and spread the knowledge on sustainable buildings. Group 4 includes governments at different levels, who are responsible for publishing regulations and voluntary agreements.

Customer Needs that are being Met and Main Benefits for the Target Clients

Building materials are mainly bought by construction companies that may work with design and/or architect studios for the design of their project. All these companies are often indicated as contractors, most of the time being appointed by an investor for the realization of a project. Building materials are also bought, to a minor extent, directly by building owners for retrofitting or refurbishment.

Building Owner	Building Tenants
Increased rental rates	Lower operating costs
Attracting tenants	Healthier, cleaner indoor environmental quality
Cost effective	Better performance
Competitive differentiation	PR and community benefits
Risk mitigation	Higher productivity

In our business strategy we have identified both contractor companies and building owners as target customers. These two different commercial targets have slightly different needs and are motivated by different reasons to buy. We have tailored MOGU floor to satisfy the needs of both our target customers (reported in Table 6) and we provide them with the following benefits that are unparalleled on the market:

1. Construction and design companies

For contractor companies, like construction and design enterprises, the selection of the flooring solution is mainly aimed at attracting new clients and maximizing the ROI of their projects.

In this regard MOGU floor offers a number of advantages to meet the needs of such target customers, including:

Increased ROI thanks to higher willingness to pay of customers:

For contractor companies, offering green certified projects has become the number one strategy for staying competitive, maximizing operative margins and reducing the payback period. Investors and owners are aware of the greater value of green building (up to 19.2%)[16] and are ready to pay higher prices for green certified projects. For example, office buildings are generally classified into three categories: Class A, Class B, or Class C.Class A represents the highest quality buildings in their market and are usually green certified. Class A acknowledgment resultsinlease-up rates typically 20% above average. In addition, in a Nielsen global survey on corporate social responsibility, more than half (55 %) of end users said they are willing to pay extra for products and services produced or offered from companies that are committed to positive social and environmental impact. Regionally, respondents in Asia-Pacific (64%), Latin America (63%) and Middle East/Africa (63%) exceed the global average. While there has been an increase in the willingness to pay extra for sustainable products are in North America (42%) and Europe (40%) in recent years.

MOGU floor will be compliant with the most important green certification labels (table 7) of the building sector and as such will bring high value to the projects of construction and design companies applying for green certification.

Reducing the payback period by attracting more clients seeking the advantages of resilient flooring:

In addition to green certification, MOGU floor will further attract clients for contractor companies, satisfying the market demand for resilient luxury flooring. As seen, the resilient flooring market is indeed the fastest growing segment of the flooring market, with a CAGR more than double than the average of the overall flooring market (10% vs 4.4%)[14]. We designed MOGU floor to be sustainable but also to deliver a warm, delightful and carefree living experience. By coupling unique aesthetic designs with durability and ease of

[16] McGraw Hill Construction (2010). Green Outlook 2011: Green Trends Driving Growth.

Reduced investment costs and simplified installation:

MOGU floor is positioned in the high tier of the flooring market and competes favourably with other luxury solutions. Typically these solutions generate high operating margins, but at the same time require high investment. Not only does MOGU floor offer greater value than the competing solutions, thanks to its sustainability, performance, ease of use, functionalities and design, it is also cost competitive.

For example the price of Luxury Vinyl Flooring (LVF) ranges between 50-85 €/m^2. As we said, these products are not environmentally sustainable, are potentially hazardous, and can only mimic a natural texture and feel. MOGU floor on the other is eco-friendly, safe and offers a natural design for a target price of 60-70 €/m^2. In addition MOGU floor does not require slab preparation or humidity testing. Moreover, taking into account installation costs, the price of LVF ranges between 80-110 €/m^2, i.e. more expensive than MOGU floor, which costs 80-90 €/m^2. In short, the price competitiveness of MOGU floor represents an important benefit for contractor companies, who will be able to propose a superior and cheaper flooring solution in their projects.

Table 7. Green product certifications

Label	Description
Blue Angel	The Blue Angel was initiated by the German government and awarded by an independent Jury to products that are environmentally friendlier than others serving the same use. The Blue Angel promotes the concerns of both environmental protection and consumer protection. It is awarded to products and services that are particularly beneficial for the environment in an all-round context and also fulfill high standards of occupational health and safety and fitness for use.
Cradle to Cradle	The Cradle to Cradle Certified(CM) Products Program provides a company with a means to demonstrate efforts in eco-intelligent design. Cradle to Cradle Certification is a third-party sustainability label that requires achievement across multiple attributes. This certification program applies to materials, sub-assemblies and finished products. Cradle to Cradle Certified is a certification mark sublicensed.
Indoor Air Comfort	"Indoor Air Comfort" product certification is an innovative tool for showing compliance with low VOC emission requirements from construction products and furniture of all relevant European specifications on two levels: Standard level "Indoor Air Comfort - certified product" shows compliance of product emissions with all legal specifications issued by authorities in the European Union. Higher level "Indoor Air Comfort GOLD - certified product" shows compliance of product emissions with the relevant ecolabels and similar specifications in EU.

2. Building owners and tenants

The second target customer of our business strategy are building owners and tenants. Typically when this category of customer buys building materials, it is as part of a retrofit or refurbishment. For this reason, they need a solution which is easy to install and that can improve the indoor environment in terms of livability, safety, productivity, design, productivity, performance and sustainability. Of course price- and value for money- is a key parameter affecting the decision making of this type of customer.

Simple solutions that can be easily installed and that do not require burdensome maintenance:

MOGU floor is a modular solution for flooring that is made of tiles, with angles shaped to provide a perfect joint that requires no welding and always provides for a seamless fit. Modularity results in quick, easy and inexpensive installations that can easily fit virtually any type of setting. MOGU floor does not require any particular ground preparation prior to installation. Unlike other solutions on the market, such as linoleum or rubber, MOGU floor can also be installed directly by an end user with minimal experience in do-it-yourself (DIY). Moreover, in the event of damage, a MOGU floor tile can be removed and replaced within few minutes. This is not possible in case of linoleum, for example, which is applied as a unique solid sheet on the floor and therefore cannot be repaired. The high level of reparability of the MOGU floor is another favourable feature in the context of the Circular Economy. Finally, MOGU floor is a robust material coated with an advanced treatment that makes it resistant to shocks, fire and water. The lifetime of MOGU floor is slightly shorter than PVC, but in line with linoleum or rubber and longer than bamboo.

All in all, MOGU is the ideal solution to meet the booming market demand for resilient, easy to install and carefree flooring. The modular design of MOGU floor does not require any slab preparation or humidity testing, allowing the end user, even with little DIY experience, to install it autonomously. Once installed, MOGU floor is resistant, durable and easy to clean allowing the end user to simply focus on their living experience.

Safety and Indoor Air Quality:

Green buildings may also reduce the risk of lawsuits over mold-related and other health issues. Since most U.S. citizens, Europeans, and city dwellers globally spend about 90% of their time indoors, the quality of the indoor environment is extremely important.Unlike the directly competing LVF, MOGU floor is 100% natural, and with a minimum VOCs released in the indoor environment, results in great IAQ. In fact, our aim is to obtain premium IAQ certification for MOGU floor, which will increase the value of the product for the end user. As IAQ is one of the parameters evaluated in the main green building certification scheme, MOGU floor will also be compliant with green building standards.

Increased Productivity and Worker Satisfaction:

The owners of commercial buildings need to refurbish their offices from time to time. Indoor environment quality has a significant impact on employee productivity, and green buildings offer better experience and IAQ for occupants to enjoy[17]. These features of a healthy work environment help to attract and retain employees.Moreover, occupant comfort and satisfaction reduce sick time and improves productivity. In fact, according to "The Impact of Office Design on Business Performance", it has been estimated that "a 2-5 % increase in staff performance can cover the total cost of providing their accommodation". That same report states that "differences in productivity as high as 25% have been reported between comfortable and uncomfortable staff". As an example, at Genzyme's headquarters in the U.S., several environmental features have made employees happier and more productive: sick time was reduced by 5%, 88% of employees reported improved wellbeing, and 72% of employees reported improved alertness and productivity[18].MOGU floor can enhance the environment of refurbished buildings by increasing staff comfort through an improvement in walkability and reducing slippages, minimizing floor-to-ceiling temperature stratification, controlling noise and improving IAQ. Additionally, individuals often benefit psychologically from aesthetically pleasant environments and from knowing they are walking on a 100% natural material.

Original and beautiful designs that improve the living experience:

Building owners investing in a refurbishment project have expectations for improved aesthetics of their building. The various designs and tile sizes potentially achievable with MOGU technology offer plenty of design possibilities of customization in terms of colour, size and shape. The use of different types of resins and the addition of natural compounds such as coffee, coconut shells or wood, infuse a unique sense of texture and a granular feeling in MOGU floor. Such naturalness and tactility respond well to the demands of residential applications for a warm and familiar feeling. For the first time ever, MOGU floor couples the advantages of resilient flooring with natural, warm and glamourous designs that are fully sustainable.

High performances and reliability:

Among the reasons for refurbishing a building, is the need to comply with new regulations or reduce operational costs. MOGU floor outperforms the competing solutions at multiple levels including fire resistance, thermal insulation and noise absorption (according to EN standard in table 4). In addition, it is resistant and shock-absorbing.

Sustainability and natural raw material:

Already in the resilient flooring market, a number of natural solutions are available, such as linoleum, rubber and bamboo. MOGU floor,being produced from residues from agriculture or industry, surpasses the sustainability of these products, making it the most sustainable solution for flooring applications.

Affordable price and great value for money:

Taking into consideration the entire lifecycle of flooring products, including installation and maintenance, MOGU floor is extremely well positioned in terms of price considering the functionalities and quality that it delivers to end-users. The great advantage of the MOGU products from an economical point of view, is the savingsmadeduring the installation and indeed along the entire lifespan. It is clear that the MOGU solutions will deliver unprecedented value for money for customers, offering top class products for performance, functionalities, design and sustainability.

MOGU floor positioning in the flooring market

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[17] Delmas, Magali A. and Pekovic, Sanja (2012). Environmental standards and labor productivity: Understanding the mechanisms that sustain sustainability.

[18] Green Buildings Pay: Design, Productivity and Ecology, Brian W. Edwards, Emanuele Naboni 2013

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Today, the resilient flooring market is populated by the following typesofmaterials:

Linoleum: a natural based material made from dried and milled flax seeds mixed with other plant material (pine resins, wood flour, ground cork) with a jute backing. It is fire-resistant and does not require an additional fire-retardantsfinish. However, linoleum is characterized by a burdensome installation, the impossibility of repairing damages and by a very unpleasant aesthetic.

Rubber:a general-purpose flooring, mainly made from a copolymer of styrene and butadiene (two fossil, nonrenewable compounds), and diffused in commercial and high-traffic applications. It is easy to install and provides effective sound insulation and vibration reduction. However, it is prone to staining, subfloor infiltration and slippage. Lately texture printing of the surface opens its application also to domestic environments, but the most attractive design starts at 50 € per m^2 and can also exceed 70 € per m^2.

Vinyl: flooring made from polyvinyl chloride.Available since the 80s and it is known as a cheap and durable solution for commercial, high traffic applications. In recent years new technological solutions have allowed then ability to realistically mimic the appearance of natural materials thanks to a printed layer on the surface of the vinyl flooring. These solutions represent a "revolution" in the resilient flooring sector and are leading to an outstanding market success of the luxury vinyl flooring (LVF) segment.As described the negative environmental impact of Vinyl along its lifecycle and the release of harmful VOCs in indoor environments are limitations of this type of resilient flooring.

Bamboo: resilient flooring can be made from natural material such as bamboo. These solutions are considered the most exclusive solutions for resilient flooring in virtue of their quality, aesthetics and exclusivity. However their cost and maintenance are far more expensive than the other flooring segments.

Tab 8: MOGU floor compared with competitors

Characteristic	MOGU floor	Linoleum	Rubber	Luxury Vinyl	Bamboo
Natural	Yes	Yes	No	No	yes
Indoor Air Quality	High	Medium	Medium	Medium	high
Fire resistance (EN 13501)	C	B-C-D	B-C-D	B-C	C-D
Durability (years)	30	25	25	40	30
Thickness (mm)	10	2.5	3-4	4-8	10-15
Noise Absorption	High	Low	Low	Low	medium
Shock absorption	High	Low	medium	medium	low
Thermal Insulation	Medium	Low	Low	Low	Medium
Reparable	Yes	No	Yes	Yes	yes
Need of MVER test	No	Yes	No	No	No
Slab preparation	No	Yes	Yes	No	no
Material cost (€/M^2)	60-70	15-40	30-70	50-90	90-120
Installation cost (€/M^2)	30-40	45-65	40-50	40-50	40-50

In addition to the type of material, many other parameters differentiate high quality resilient flooring from inferior solutions, including: thickness and size of tiles/sheets, wear layer, quality of register embossing, quality of visual layer, edge treatments, customization options, sustainability and warranty. On the basis of these characteristics, the resilient flooring market can be divided in 4 segments: entry/low, mid, luxury and premium.

The entry segment consists of the cheapest solutions for industrial environments, such as traditional linoleum and rubber flooring.

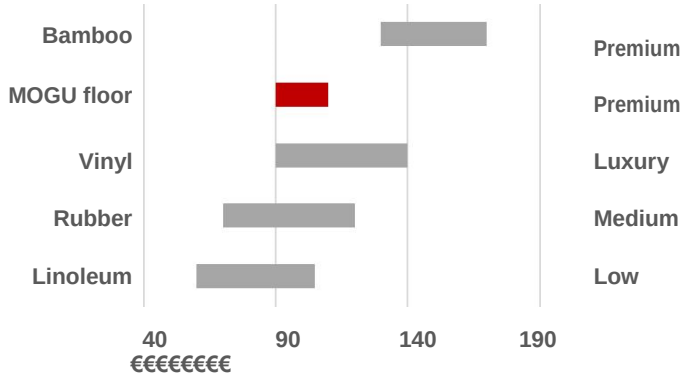

Fig 5: Materials and segments of the resilient flooring market

In the mid segment, we canfindnew products based on linoleum, rubber or vinyl. They are slightly more expensive than the previous ones and typically designed for commercial applications. The luxury segment is the one driving the growth of the resilient flooring market, thanks to the success of LVF. Modular products with natural-mimicking printed surfaces are sold in this segment. Finally, the premium segment consists of the most expensive and exclusive products, such

as bamboo, that provide authentic natural patterns and texture.

We have designed MOGU floor in order to deliver end users the possible highest value for money. MOGU floor excels in thickness, shock and noise absorption. It is durable, fire resistant and safe. While some of these characteristics can also be found in some of the other products in the luxury segment, MOGU floor has the added advantages of being 100% biobased and sustainable and offering a real and natural texture that far exceed that of printed products like rubber or vinyl flooring. In short, it is clear that MOGU floor outperforms the products in the luxury segment (see table 8) and competes directly with the premium segment of the resilient flooring market. However, a clear advantage MOGU offersoverthe premium competitors, is the selling price, which we envisage to be typicallycomparable to that of lower segments, as shown in figure 5.

Main Competitors and advantages of MOGU floor

As described, MOGU floor will occupy the premium segment of the resilient flooring. This is a very favorable positioning as it is the fastest growing and most profitable segment of the entire flooring market. The cost effectiveness of MOGU floor will make it a product with probably the highest quality/price ratio on the market.

Tarket: Veneto, Etrusco, Trentino, ID Inspiration, ID Square (Italy)

Tarket is one of the leading companies worldwide in the resilient flooring market, offering several product types, characterized by different feedstocks and technology. In particular Tarket provides luxury vinyl tiles (ID series) with different finishing and thickness. Tarket is also one of the main linoleum producers (15 products, manufactured for 150 years), mostly targeted at the commercial segments. The linoleum flooring is treated with a unique surface protection for extreme durability, easy cleaning and cost-effective maintenance. They also have a niche flooring that provides high acoustic performance, appreciated in the hospitality sector.

Nora: Norament, Noraplan (Germany)

Nora Systems GmbH produces and markets resilient rubber floors under the brand name Nora. Norament tiles offer a classic rubber floor covering, with an interesting combination of quality and functionality thanks to a high-pressure process that produces very dense tiles resistant to wear supporting fast and easy cleaning. In our tests, MOGU floor reached similar performances but with a far lower density, allowing resource savings and lower production costs.

Forbo: Marmoleum, Acoustic linoleum, Allura (Spain)

Forbo is a global provider of premium commercial and residential floor coverings, with an extensive range of environmentally friendly natural linoleum, luxury vinyl flooring and carpet tiles. Marmoleum series are linoleum solutions, with more than 12 different design structures to choose from, available in sheet formats. Allura lines are luxury vinyl tiles designed and produced in Europe, with interesting dimensional stability so that the appearance of the floor remains as seamless as the day it was installed. However, none of these two product lines reach the thickness or texture and feel quality of MOGU floor, as their surface is printed and not real.

Mannington: Adura and Distinctive Plank (USA)

Mannington's luxury vinyl planks are divided in two collections Distinctive Plank, and Adura.The latter is the cheaper option with 4 mm thicknesses. The Distinctive Plank line includes full 15 x 120 cm sizes, micro-beveled edges, and darker, more soothing colors than those found in Adura. In addition, Distinctive's embossing is far more complex and true-to-life than Adura's, but is still just 4 mm thick. MOGU floor offers a more flexible modular design and higher quality tiles.

Commercialization Risks and Market Barriers to Overcome

At MOGU, we are fully aware that in addition to the great advantages and USPs of MOGU floor, there are also a number of potential risks. A solid business strategy must consider multiple factors including the following risks and market barriers:

Secure the supply chain: MOGU floor is made starting from the leftovers of other processes. For this reason, raw material availability could be affected by fluctuations that could limit the production capacity of MOGU floor. To totally avoid this risk, we already have a supply contract with CFT Masserini (Bergamo - Italy).However in order to diversify our suppliers and extend the range of customization of MOGU floor,

during the SMEIPhase 2 we will identify additional new suitable suppliers/residues Associatedwithdocument**and**Ref**arrive**.Ares(2018)3790472**atcommercial** - 17/07/2018 **agreements with them.**

MOGU floor certifications: The CE mark sets the basic technical requirements to be met in order to obtain a marketable product. It is the first certification that MOGU floor needs to acquire, as specified in European Construction Products Regulation CPR 305/2011/EU. The CE mark is, from a legal point of view, the only certification needed to sell MOGU floor. Nevertheless, limiting the certification of MOGU floor to the CE mark would represent an under valorisation and would result in a commercial failure. We will obtain additional certifications that will increase the value of our product on the market and the willingness to pay of our potential clients. Given that there are a great number of "green" certifications available for building products, in order to ensure we focus on the most valuable once for MOGU floor, we included a dedicated task in the workplan (Task 1.1)in order to allow us to review and evaluate which ones to apply for.

Distribution Channels: We are a new player in the resilient flooring market and as such we need to identify the most effective way of reaching the end users with our products. From this context, the competitors already active in the market have a dominant position. To disrupt the current market scenario and to rapidly gain market share, as previously described, we will focus our strategy on collaborating with the most influential stakeholders from the building market. Both construction companies and building owners are directly or indirectly affected in their decision making by a number of stakeholders, who are considered the reference points of the building market and who contribute to setting the trends of the seasons. At MOGU, we are already initiating several collaborations with such influential stakeholders with a view to securing privileged market access. In addition, traditional retailing and ecommerce distribution channels will be used to penetrate the flooring market.

Consumer Awareness and acceptance: MOGU products are unique in the market and their commercial success depends on customer recognition of MOGU value. While other brands are already known and can rely on a loyal user base, MOGU need to build its brand identity and gain clients. For this reason a tailor-made marketing campaign and initiatives will be deployed before and after the product launch. Our strategy of collaborating with renowned trendsetters and influencers of the interior design sector will be instrumental to building our brand.

Capital requirements: At our facility, we are currently capable of producing 50 m^2 per day of MOGU floor. This operational capacity has allowed the demonstration of our solutions to TRL 7. Larger investments are now needed to close the gap for commercialization of MOGU floor and we are applying to the SMEIPhase 2 as part of our fundraising strategy. In this context, MOGU has already secured a first-round investment, with the Italian Investment Fund Moffu Labs (1.7 M€ post-money valorization).

2.2 Business model

Alignment of the Innovation with the company´s overall business strategy

Our company goal is to become a reference player in the green building sector for innovation and design. In line with this vision, we have devised a company strategy based on developing a new technology for the production of advanced composite materials to apply in the building sector. The first product we have developed with our technology is MOGU floor. By exploiting its tremendous market potential, we aim to grow our business and reach international markets. With our technology, further materials and products (such as acoustic insulating panels) will be developed in the future to sustain the growth of the company and open up new business lines.

The SMEI represents a great and timely opportunity to accelerate our business strategy and drastically reduce the time-to-market of MOGU floor. This proposal indeed has been designed to bridge the final technology and business gaps that divide the current development of MOGU floor from the market, and to prepare the future growth of the company. The proposal indeed follows our strategy by:

Completing the development of MOGU floor: Even if our product is at a TRL7, it still needs some developments, such as the definition of our first MOGU floor collection and the optimization of the manufacturing process. In parallel with the technological tasks it is crucial from the commercial point of view to obtain the most relevant green certifications. Both these aspects are covered in WPs 1 to 4.

Deliver MOGU floor to our customers: In order to ensure we reach the end user with our product, we are devoting great effort to collaborating with interior design and architecture studios that are recognized at international level. The B2B market will be addressed in the short term (2018 and 2019) predominantly

invaluable opportunity for marketing and communication. The B2B strategy will be implemented directly by

our team, whereby we will negotiate the most appropriate type of collaboration and transmit all the values of our MOGU products.

The market strategy includes also international commercialization. As such, a dedicated international sales team will be established. This team will have geographical objectives and will develop relationships with local retailers and stakeholders. To this extent they build on our existing relationships with existing players in the flooring sector. In addition, local market specialists will coordinate the building of a network of non-exclusive agents that will be engaged from the end of 2019 onwards.

Allow sustainable long term growth and internationalization: We are focusing our commercialization strategy on building a strong brand identity in the market of sustainable building materials, consolidating our business at international level and in the future launching new products in different market segments. In this way, not only the short term viability of our business will be ensured, but there will also be the possibility of further growth in the mid-long term thanks to the extension of our platform technology to other applications. Indeed, we are already working on other product prototypes at lower TRLs (3-5) in the automotive and packaging sectors, which can be commercialized in the future, taking advantage of the market positioning achieved with the commercialization of MOGU floor, as well as leveraging interest from important players as Novamont (letter of support in section 4-5).

Our business model and the value chain for ensuring successful commercial exploitation

We have devoted tremendous effort to developing a technology beyond the state of the art, and our business model has the potential to disrupt the current market dynamic, and will be instrumental to scaling our business.

We have redesigned the traditional value chain of the building sector starting from the leftovers of other processes to create a fully biobased product. Notwithstanding the new policies and political efforts to make the building sector more sustainable it continues to be predominantly based on the "take-make-waste" linear business model. At MOGU, our ambition is to change this unsustainable approach and bring the Circular- and Bio-economy to the building sector. The environmental and social benefits of such transition are also a great business opportunity at multiple levels.

In the first instance, we are able to obtain the raw material needed for our MOGU floor at a competitive price. This represents a great competitive advantage over linoleum or vinyl flooring that use as raw materials flax seed and PVC that cost 300 and 400 €/ton respectively.

Moreover, in comparison with synthetic rubber or vinyl, MOGU floor does not rely on fossil resources and therefore is not subject to price fluctuations due to geopolitical dynamics. At the same time MOGU floor is not produced from dedicated non-edible crops (such flax seed for linoleum) and therefore does not compete with any crops for lands and is not subject to climate phenomena such as flooding or drought.

The use of byproducts from other industrial processes result in the valorization of other value chains that do not need to cover the cost of disposal but on the contrary can generate additional incomes.

Our business model is centered for the most part on B2Bsalesto constructors and other related companies from the building sector, and to a minor extent directly on B2C sales directly to end users.

The B2B commercialization of MOGU floor will be focused on a network of collaborations with influential stakeholders of the building sector, as already described. This will enable a faster market penetration and gain of market share, which could otherwise represent a market barrier for a new player such as MOGU. B2B will be performed on the basis of supply contracts for relatively high amounts of product.

Regarding B2C sales, we will use exposition space of existing resellers in strategic locations, and such an agreement has already been reached with ExT.Part of the B2C sales will also be secured through e-commerce. As discussed, even if still a minority of sales in the building sector are performed online, this is an area that is experiencing fast growth. Improved website functionalities and future technologies, such as for example augmented reality, willseeecommercebecome a prominent platform for thesaleofbuilding products.We will develop an ecommerce portal that will also inspire our customer with high quality images of our products and application examples.

Commercialization plan with an approximate time-to-market or deployment, and international scaling plan

Assessed with document - 17/07/2018

To enter in the targeted market segment (luxury resilient flooring) and reach our customers, we have developed a comprehensive commercialization strategy with the support of our Advisory Board and Mentors (see Section 3.1 Team) as part of the acceleration programs that we have been awarded places on. As a result, a three phase commercialization roadmap has been defined:

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Phase 1: Preparation of market uptake

We expect to be ready to commence commercialization of MOGU floor by M18. In the 18 months prior to this we will use this period to prepare for market uptake by: 1) increasing the production capacity of our plant; 2) obtaining CE mark and addition green certifications; 3) promoting MOGU floor and rising the hype around its market launch; 4) signing commercial agreements and preparing distribution channels for selling MOGU floor.

The plan for upscaling the production plant and for acquiring all the certifications are described in the workplan in Section 3.2 (WP3 and WP4 respectively). For preparing the market uptake of MOGU floor, we will focus our strategy on increasing visibility for MOGU floor, and especially in prestigious channels, on communicating its Unique Selling Points (USPs), and in general on leaving a positive imprint on the decision making process of our customers.

For achieving this result we plan to participate to the most relevant events in the building and material sectors, publish articles in magazine, blogs and papers and raise the community interest through social media campaigns. Importantly, we will to make MOGU floor available in International Material Libraries. Stakeholders of the building sector can access Material Libraries to discover new materials and stay up to date of the latest trends of the market. Material libraries are collections of samples of the most innovative and glamourous materials, that are selected by experts and exposed in galleries. It is noteworthy that MOGU floor is already available in the Material Libraries of Materfad (Barcelona), Materialbiblioteket (Stockholm) and soon will be available also in Material ConneXion (NYC), Parsons School of Design (NYC), IMAT-UVE (Moncheengladbach), Materiò (Paris), Matrec (Milan). The presence of our product in such Libraries is a great opportunity for reaching and attracting new clients with virtually no investment. Moreover, being a prestigious showcase for MOGU floor, our presence in these libraries can be used as a very powerful marketing tool.

Phase 2: Market penetration driven by early adopters

While the actions of the phase 1 of our commercialization strategy are an invaluable means to increase awareness about our product and its benefits,the actual conversion rate of these activities into sales can be low. As such, to increase our market penetration and customer adoption, we plan to engage the most influential stakeholders of the building sector as early adopters of MOGU floor and leverage these collaborations to gain market share.

Architects, designers, planners, engineers, consultants and technical specialists are involved as advisors or contractors in the largest part of any building project, and especially in the most prestigious and valuable ones. Such stakeholders have a deep influence on market trends since, as their opinion directly influences the decision-making process of their clients. They also have a far reaching indirect influence over all the end users that will look at their projects as references and examples of the new trends of the building market.

Such stakeholders will be tapped into as early adopters of MOGU floor in order to gain fast access to the building market and quickly build market share and popularity. We already have early adopter agreements for MOGU floor with influencers such as ARUP, a multinational firm which provides consulting services for all aspects of the built environment across 42 countries, and ATI Project, a team of specialized experts in the field of architectural design with projects in more than 20 countries (see the Letters of Support attached to this bid). Taking advantage of the SMEI support and visibility, and starting from M4 (once we have defined the first collection of MOGU floor), we aim to further extend our collaboration network with strategic stakeholders that are ideally positioned to driving the market demand and generating hype for MOGU floor. Indeed, such stakeholders are constantly on the lookout for innovative new products to include in their projects and that offer great value and cost competitiveness. MOGU floor represent an innovative product, which is unparalleled on the market and is capable of beating competing solutions across many levels of performance, look and feel.

As of M18, we will be in a position to supply the market with MOGU floor in volume and as such we will open new distribution channels for reaching a wider diffusion of our product. We already have an agreement 20

France and Spain, to commercialize MOGU floor in their stores. Through WP5, we will work towards signing more agreements with national retailers such as Liuni to further increase MOGU floors availability on the market.

In parallel to traditional distribution in physical stores, we will develop an ecommerce portal for showcasing the MOGU floor experience to perspective clients and facilitating enquires for product samples. Even if traditional distribution channels are still dominating the sales of the flooring market, over the last number of years eCommerce is quickly growing. According to the U.S. Census Bureau of the Department of Commerce, e-commerce sales of flooring in the first quarter of 2017 accounted for 8.5% of total retail sales. That compares to 7.8% of total sales in Q1 2016, 6.9% in Q1 2015 and 6.2% in Q1 '14. Five years ago, e-commerce represented 5.1% of total retail sales.

Phase 3: Internationalization

International sales are one of the main objectives of our business strategy. As such, after 18 months from the end of the SMEI Phase 2 project, we intend to expand our sales at international level. According to the market analysis performed to date, we have selected Benelux (market size of 0.65 B$), Germany (3 B$), Russia (0.7 B$) and Scandinavia (0.4 B$) as the initial target markets. We will certify MOGU floor taking in consideration the more recognized and valuable certifications for our target markets. For example, in Germany the Blue Angel label, in Scandinavia, the Nordic Swan (officially recognized in Finland, Sweden, Norway, Iceland and Denmark), in Benelux, the Milieukeur (the Dutch environmental quality label), and finally for Russia international labels such as BREEM (already described) and EcoMaterial (certifying that a material achieves high performance in human and environmental safety and indoor environmental quality and raw materials selection) are recognised. As a proof of the importance of green labels in these countries, some of our competitors have products specifically certified for these markets, such as the linoleum flooring by Forbo which is Nordic Swan certified and the rubber flooring of Nora which is Blue Angel certified. To gain visibility, a better understanding and to start commercial relationships in international markets, we will attend the most relevant industry events and trade fairs (e.g. Domotex, Nordbygg, Formex, Fair living).

Market entry in these target countries is scheduled for 2019, with a significant ramp-up in the next three years. During this period we will work with local retailers and non-exclusive agents for the promotion of our products. In particular, in 2019 we aim to commercialise MOGU floor in the Scandinavian countries and to achieve a significant penetration of this market. It is noteworthy that we have already hired Olga Volkova, a skilled commercial manager, based in Sweden. Commercialization in Germany, BeNeLux, Scandinavia and Russia is scheduled for 2020.

On the longer term, two additional actions are planned for boosting revenues from international markets and starting from 2021. The first one is the licensing of MOGU floor IP to third parties operating in sectors not competing with our commercial activities. While for us it is capital to keep and reinforce our USPs for the resilient following sector, our natural grown material can also be used in other applications in the building market, as well as in other sectors such as automotive, nautical, horticulture (biodegradable pots) and packaging (rigid).

2.3 Financing

Company's ownership and capital structure

In 2018 Mogu share capital is 24,435,€, with shareholder structure composed as follow: Stefano Babbini (46%), Maurizio Montalti (28%), Federico Maria Grati (9%), Francesco Amadio (9%), Francesco Giannetti (9%), Moffu Lab Srl (5%). In December 2017, on the occasion of Moffu Labs partnership, a financial due diligence has been conducted, consolidating a company asset of about 140,000 €, and, taking into account the business plan and intangible assets, a company value of 1,7 M€.

Impact of the innovation on the company financials (profit/loss, turnover and cash flows), jobs and efficiencyproductivity improvement

With the support of the SMEI Phase 2, we are aiming at reaching a 0.05% share of the European resilient flooring market share in five years. This ambitious target corresponds to a production of 180,000 m^2 of MOGU floor per year. To achieve this objective our business plan envisages a time to market of 18 months from the beginning of the project. The roadmap to commercialization includes the definition of MOGU collection (MS1 at M4), the design of the upscaled production plant (MS3 at M10), the achievement of the

the SME I project) it will be launched in: Netherlands, Germany, Sweden, Norway, Finland and Denmark. To enable this growth and to scale our team, a recruitment campaign is expected to sustain the production capacity and reinforce the commercial unit.

We have estimated that the implementation of our business strategy will result in a cumulative turnover of approximately 35 M€, in addition to the creation of at least 50 direct and 20 indirect jobs.

The recruitment plan includes the hiring of a Human Resource specialist within the first 6 months of the SMEI project. From there we plan to hire an additional team member in the first 12 months. In the second year we will hire 2 market specialists, 1 controller, 2 technicians (electrical and mechanical), and 20 workers. In the third year a CFO will be contracted, together with an office manager; the manpower will increase as well (10 persons). In the fourth year the technical capability will increase, hiring a mechanical engineer, and a material scientist. Finally in the fifth year the manpower will reach the steady state (30 persons); a new market specialist will be contracted as well. A 5 years financial projection is in table 8.

Table 8- Growth projections for 2018-2022

	2018	2019	2020	2021	2022
Revenues	270,000	5,475,000	7,195,000	9,660,000	12,000,000
Sales of final product	270,000	5,400,000	7,020,000	9,360,000	11,700,000
Licenses	-	50,000	100,000	150,000	100,000
Sales of MOGU feedstock	-	25,000	75,000	150,000	200,000
Costs	1,076,529	3,898,035	5,553,007	5,396,621	5,652,226
Fixed general costs	55,000	76,000	96,000	100,800	105,840
Industrial production (feedstock)	35,114	573,798	861,810	387,624	478,690
Industrial production (final products)	153,915	2,084,237	3,100,197	3,100,197	3,100,197
Labour	729,000	934,000	1,219,000	1,509,000	1,634,000
Management	103,500	230,000	276,000	299,000	333,500
EBTDA	- 806,529	1,576,965	1,641,993	4,263,379	6,347,774
Net profit	-299%	29%	23%	44%	53%
Margin	597,904	916,501	914,507	2,662,796	4,051,300
Capex	699,982	1,231,878	680,042	607,004	20,000
Working capital	- 169,232	-262,762	- 281,940	376,997	352,015
Cash Flow	- 1.381.253	- 369.086	229.582	2.721.846	4.684.372

As a valuable and innovative solution in the flooring market, the MOGU business plan is highly attractive for the leading companies operating in the green building and bioeconomy sectors at EU and worldwide level. Such a scenario represents an exit strategy opportunity, which has been positively evaluated by potential investors.

Funding Requirements for Reaching International Commercialization

Our fundraising strategy has been thoroughly elaborated in collaboration with PTP Science Park, UnicreditStartlab Academy, and Startup Initiative of Intesa San Paolo Bank, during three acceleration programs won in 2016 and 2017. The strategy has been developed along 3 main steps, aligned with our industrial plan: product development (seed and angel investment), market penetration (VC investment and institutional support - SMEIPhase 2), full commercial scale (private equity investment accompanied by project financing and/or bank loan leverage).

The first funding stage has already been secured during 2017, with the investment of the co-founders and Moffu Labs (100.000 €). In particular, the Investment Fund Moffu Labs Srl, based in Milan (Italy), made an angel scale investment in MOGU acquiring 5% of shares, and recognizing a valorization of 1.7 M€ post-money. These resources have been used for the expansion of our pilot facility up to the productive capacity of 50 m^2 per day, the development of MOGU floor prototypes and the first promotional activities. With Moffu Labs, we have also signed a consulting contract, where Moffu Labs is acting as a business and commercial advisor and in particular in relation to the company market entry and the industrial scale up in the medium term.

The second stage of our fundraising strategy opens in 2018 with the objective of preparing and launching the commercialization of MOGU floor at national and international level. The SMEI has is linked to two pivotal

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objectives in our strategy: providing us the resources to bring MOGU floor to market and leverage additional investments thanks to the increased visibility and reliability. We are already in contact with potential investors, such as Invitalia Venture, the Italian Government early stage VC which showed interest in our technology and our business plan. To promote MOGU and attract more investments, we are also actively pitching our company in the most important startup events and roadshow, such as the BioinItaly, Startup Initiative (in collaboration with Ellen McArturFundation for the Circular Economy), Biofabricate (NYC- USA), UnicreditStartLab investments arena, Klimahouse startup arena, World Forum for Responsible Economy (France), Bioeconomy Startup Summit (Germany).Thanks to our proactivity we have received several Expressions of Interest from corporate funds (mostly from the chemical and biotech sector) and private investors. Among them a group of private investors leaded by Incatro owners, are evaluating the support to

the development of an additional production plant in The Netherlands, ideal for the market expansion in Benelux and Germany, two of our target markets.

Finally, the third step will consolidate MOGU positioning in the European market and open new commercial channels in Asia and USA. This stage will be partially financed by reinvesting the revenues of our commercial activities and by additional funds from private investors. This phase could alsobenefit from the support of the SMEI Phase 3.

2.4 Intellectual Property Right (IPR) and legal framework

Legal and regulatory requirements to be fulfilled

The marketability of building materials in Europe is regulated by the Construction Products Regulation (CPR) 305/2011/EU, which lays down harmonised rules for the marketing of construction products. For being market ready, MOGU must be granted with the CE mark to ensure compliance with the CPR. This procedure has already been studied by MOGU, the preliminary data needed to fulfill the application has been prepared and the list of accredited certification bodies in Italy has been selected and offer proposals for certification services have been requested.

Current Industrial Property Rights Assets

The MOGU brand is trademark under registration (the preliminary assessment has been undertaken by the attorney PGA Intellectual Property). We will also register MOGU floor as a trademark. We verified the possibility to register both trademarks have been performed with positive results and the application for registration has been filed.

The technological knowledge of MOGU can be divided in two fields: the biotechnological knowhow to select and grow the fungal strain to convert the raw material into the intermediate product, and the engineering expertise for its processing into the final product. The selection of fungal strains has been developed in collaboration with the University of Pavia and is covered within the framework of an exclusivity contract ruling the supply of the fungal strains. The terms of the contracts appoint MOGU as the only commercial beneficiary of the activities developed at University of Pavia. Also strict confidentiality conditions apply to University of Pavia, whereby they must inform MOGU prior to any publications relating to the fungi strains development, and can only publish upon receipt of our permission so as to avoid any leakage of confidential information.

The knowhow for the production of the final products is entirely owned by MOGU and protected with industrial secret. Only 3 people within the company know the entire production process conditions and all employees have signed non-disclosure agreements to prevent the diffusion of confidential information. The knowhow for the production of floor includes the growing conditions of the fungi, the conversion of the material into the final products and the customization of the equipment to produce the tiles, which have been tuned by MOGU engineers.

Knowledge Management and IP Strategy

Creating a strong IP portfolio offers numerous advantages: it affords us the possibility to gain a strong positioning, both at market and technological level; increase the company value and foster fundraising; prevent competitors entering our market; and increase our revenues by licensing our proprietary technologies or products.

To leverage these advantages, we will first of all increase our brand identity and product recognizability. As described, the MOGU floor name and logo will be registered as a trademark.

the unitary industrial design right that covers the European Union (Commission Regulation (EC) No 2245/2002). According to the definition of design "the appearance of the whole or a part of a product resulting from the features of, in particular, the lines, contours, colours, shape, texture and/or materials of the product itself and/or its ornamentation". MOGU floor satisfies the protectability criteria in aspects such as material, texture and colors.

In parallel with the reinforcement of the brand and product identity, we will also protect the technological aspects of our process and product. In relation to the outcomes of the SMEIPhase 2, it is expected that the following elements of MOGU floor will be protectable:

MOGU material:Even if many mycelium materials have already been described, MOGU material is a unique composite material resulting from the synergy of the biological activity of fungi and the technological processing including different drying, molding, coating steps and the addition of resin. The result is an upgraded material,which to the best of our knowledge,is unparalleled by any other materials on the market.

MOGU technology: The process for the production of MOGU material involves several biotechnological and engineering solutions that are novel to the state-of-the-art. In particular, thanks to the optimized and specific process employed, the pressing phase, the resin casting and the related automations, all show opportunities for further IP. Therefore, the feasibility of protecting the manufacturing process willalsobe evaluated.

Genetic IP: In collaboration with a team from the Microbiology Department of Utrecht University conducting experiments on mycelium genes, we have discovered improved characteristics of fungal strains. A patent application has been filed in January 2016. Thanks to the existing partnership, we are finalizing the negotiations of an exclusive option for the acquisition of the IP. Our strategy is to use different IPRs to create a strong brand identity to increase our company value, and to protect our technology from competitors in the building sector, while licensing its use in other sectors to take advantage of royalties.

Freedom to Operate

MOGU has already carried out the Freedom-to-Operate (FTO) analysis in collaboration with our patent attorney, PGA Intellectual Property.

Research related to the utilization of mycelium as a binding element for the production of bio-materials started in the early 90's thanks to the work done by the Japanese scientist Shigeru Yamanaka, who was exploring the viability of a range of products (paper, building materials and heat insulating materials). Yamanaka, together with Kikuchi, patented the production of a composite material of fibers and fungi as binding element has been described already in 1990 "Complex of fibers and fungi and a process for preparation thereof - US 5074959 A". This patent is now expired and just represents prior art; it does not affect our FTO. No other IP for composite materials made using fungi have been identified in the flooring sector, nor has any IP claiming an upgrading effect on the raw material similar to the one obtained by MOGU floor. In our product, the organic feedstock is upgraded by the synergistic effect of the fungi metabolism and the technological processing (drying, pressing and resin casting), which results in improved mechanical and physical properties.

Even if currently we have the freedom to operate, we will update the FTO analysis every six months as best practice, as well as to confirm the validity of the results.

2.5 Communication and access to research data

Fig. 5: An example of MOGU floor installation

We have mapped out the stakeholders involved in the commercialization of ourproducts, and grouped them into different categories. These stakeholders will be the target of comprehensive communication activities that will be performed in order to successfully engage with each as discussed in turn as follows:

Beta testers and early adopters: As discussed already, close collaboration with relevant stakeholdersfromthe building sector is a success factor for our market oriented MOGU floor development. Beta testers will be iteratively involved for the definition of the product specifications, in order to develop a

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product collection including all the must-have features desired user.Among these beta testers, MOGU will collaborate with ExT for developing demo installations in selected locations, as in Fig. 7) in keeping with customer expectations.

Distributors: MOGU is aiming to commercialise its products across Europe and to reach international markets. As such building a network of distributors and sales agents its mandatory. WE are already establishing contacts with Liuni (Italy), Objectfloor (Germany), to perform preliminary scouting activities of the German, French, and BeNeLux markets and for preparingourproduct launch in these countries.

Interior design and architecture studios: As seen, interior design and architecture studios play an important role in the adoption of building solutions. Acting as consultants for construction companies and end users, they have the power to propose certain products over others, thereby deeply influencing the consumer decision making process. We will take advantage of these stakeholders to boost the market introduction of our products and we already have established relationships with important players, such as ARUP, whose Material's experts act as consultantsin relation to the selection of the most innovative, sustainable and performative solutions to be involved in large scale projects (public and private buildings, sky scrapers, compounds, etc.).Moreover MOGU has initiated discussions with the teamfromPolifactory in Milan (part of Polytechnic University), composed of designers, architects and engineers, with the purpose of promoting MOGU floor in the portfolio of solutions suggested to relevant projects. A further opportunity is represented by the network of our shareholders Moffu Labs, who are ideally positioned to link MOGU with prestigious Italian design studios, operating worldwide.

Industry associations and clusters: This group of stakeholders can provide a very effective pathway to their members. They also enable active engagement with policy makers and in helping the industry and policy-makers to work together towards common goals in relation to process efficiency, quality, safety, sustainability, etc. MOGU is already member of the following clusters: Lombardy Green Chemistry Association, Assobiotec (Federchimica - Confindustria); the Company is also joining SPRING (national cluster for the green chemistry industry), the italian Green Building Council (GBC), and the European Resilient Flooring Manufacturers Institute.

Accredited certification bodies: Engaging certifications bodies during the development of the product can simplify certification procedures ahead of the market launch. MOGU has made a preliminary screening of the main laboratories assisting in products certification in the building sector (Istituto Giordano, Certimac, Catas, ICMQ).

Renowned academics and experts: Publications in peer-reviewed journals and articles in specialized magazines can also foster the adoption of MOGU floor. We will perform a number of actions to obtain visibility through these channels, and to this end we will leverage our existing collaborations with Polifactoy Team, and Indexlab of Milan Polytechnic University.

In relation to outreach to the general public, which indeed is highly relevant for a product such as MOGU floor, we will ensure that we successfully leverage digital channels: social media, website and our e-commerce platform. Moreover, MOGU has already generated interest from among several media players (generalist and specialized), and we aim to benefit from product reviews, articles and interviews.

How we will manage/exploit/share the research data generated and/or collected during the project? Growing and processing conditions are part of the know-how on which our competitive advantage is built and on which we are going to elaborate our IP protection strategy. For this reason we are not going to be part of the Open Data Initiative.

Measures to Provide Open Access to Peer-reviewed Scientific Publications

We intend to publish in peer-reviewed Green Open access scientific publications. However, through our timely knowledge management strategy, we will ensure that no publication will prevent Intellectual Property Protection.

3. Implementation

3.1 Team

Our team, their achievements and experience,roles within the project, and the main strengths and weaknesses

MOGU has a balanced shareholder structure, composed by Stefano Babbini (CEO) and Maurizio Montalti (R&D Director), Federico Maria Grati (Member of BoD), Francesco Giannetti (Product designer) and Francesco Amadio (Product Certification and Ecolabel expert).Since its foundation, our team has had a strong and expert core, not only in technology but also in business innovation and today the Team is organized in units covering the entire expertise needed to deploy our business strategy as follows:

- Stefano Babbini - Mogu CEO - has been active in bioeconomy and agro-Industry sector since 2005. Graduated in Environmental Engineering in Florence University, he contributed to start-up and ramp-up of relevant initiatives in EU and in Developing Countries, in the bio-energy, agro-industry and green chemistry (about 100 stable jobs created). Beside the management experience developed in field, he increased his skills attending exclusive programs focused in the Business & Administration (UnicreditStartlab Academy, Alimenta2Talent). He co-foundedMogu in 2015, and today he is fully dedicated to its business development, in charge of the Company business plan and strategic decisions. He is also responsible of Mogu partnership in BBI Horizon 2020 Consortia.
- Federico Maria Grati, Board Member & Business Advisor, is an environmental engineer with 10+ years' experience in the bio-materials, and agro-industry sector. Founder of various innovative start-ups, he has been honored with BEST Fulbright scholarship and GSBI in 2011. Previously manager in M&G Group and Clariant, today he is the General Manager of Agrigeorgia (Ferrero International SA).
- Maurizio Montalti, R&D Director: is a designer, engineer and creative entrepreneur with 5+ years' experience with production of mycelium materials. Thanks to such experience, today he is responsible of all technical R&D of MOGU materials and products, coordinating also all collaborations with the Scientific Partners in Italy and The Netherlands (Utrecht University and Pavia University).

In the 5 years following the beginning of the SMEI, MOGU is expecting to growth its team with the addition of 50+ people highly qualified and experience professionals. To properly manage such scaling of the team, a specialist in Human Resources and Talent Management will be hired within the first 6 months of the commencement of the SMEI project to bring long lasting value to the MOGU team.

Technical Development and Industrial Operations: This unit covers all the technical aspects of MOGU business which are mainly related to materials, process and logistic. The main profiles that will be involved in the project are:

- Antoni Gandia, CTO: with a Master in Plant Molecular Biology (IBMCP), he has been working in the technology development of bio-materials from microorganism, with a special focus on mycelium bio-composites (experience in the US company Ecovative Design). During the last 5 years Antoni tested several growth protocols with different fungal strains for materials production, collaborating also in the mechanical post-treatments set-up. He will be in charge of the technical development of the company, from the processes to the products characteristics.
- Serena Camere, Product Development Manager: a multidisciplinary industrial designer and Post-Doc researcher working with bio-based Emerging Materials. During her studies, she collaborated with several design studios and industrial companies, such as Skitch, Serralunga, Alessi, Woodnotes, Camparisoda, Eurochocolate, Design Innovation and Fiat-Chrysler. She is in charge of products development, according to technical and design requirements, and reporting to CTO and CEO.
- Lisa Bigozzi, Director of Operations - an Environmental Engineer, reporting to the CTO, she is in charge of the process and technology scale-up to achieve an industrial set up in the existing facility. She is also involved in the MOGU partnership in EU-Consortium under BBI Horizon 2020, and she will cover the role of Project Manager.

Even if MOGU team includes relevant engineering expertise, the design of the demonstration plant and the development of the monitoring and control system has been identified as a high risk activity to be developed in house. For mitigating such risk the support of a subcontractor has been selected based on the best value for money approach (see section 4).

Market and Commercial Development: Starting from month 18 MOGU floor will be launched on the market and a strong sales force has to be in place to push the market introduction of our product. The market unit is led by:

- Commercial Director - Alessandro Forzinetti has 10+ years experience in the commercialization of building materials, with particular focus on flooring solutions. In MOGU, his knowledge of the market requests in terms of product characteristics, he supports the product development and is the Commercial Team Leader.

- **Market Specialist North EU and Russia - Olga Volkova is a market specialist, with document and project manager with 10 years experience is the development of commercial activities in the interior design and building sector in**

Baltic Countries, Germany and Sweden. She is in charge of the market strategy in North EU. Advisory Board: During the first 3 years, Mogu appointed an Advisory Board (AB), composed of senior profiles with different backgrounds, that we consult with individually for specific tasks. Besides the individual support, as of 2018, an AB meeting will be organized (the target is twice per year), with the aim of stimulating discussions in relation to the company strategy, and gaining insights and expertise for the scale-up. The AB will be renewed annually, whereby new experts will be nominated in line with the company development.

- **Technology Advisor - Han Wösten studied biology at the University of Groningen and received his PhD in 1994. Han Wösten was appointed full professor at the University of Utrecht in 2001. He is head of the Department of Biology since 2011 and the president of the Royal Dutch Society for Microbiology since 2009. He is providing support to the Team regarding microbiological aspects of the mycelium technology at technical and network level.**
- **Communications Advisor - Sara Cristaldi, economic journalist in the Sole 24 Ore Group, and in charge in Milan Municipality for Foreign Media and Expo 2015, is supporting MOGU in communication strategy, and networking.**
- **Financial Advisor - Gianluca Ponzellini – MOGU advisor since its foundation (2015), supporting the management team regarding financial aspects, and fundraising strategy Senior Accounting Manager and consultant, founder and partner of the accounting office MetodoSrl in Milan. He is an effective member of Boards of Auditors or Boards of Directors of Italian and international companies.**
- **Business and Industrial Advisor - Riccardo Schiatti.Long standing experience in the Clean Tech Industry (McKinzey, Falk-Actelios, M&G). He is also a partner at Moffu Labs, an investment fund based in Milan, with the focus of supporting technological start-up companies in the industrial scale up. The Company is composed by private investors, with excellent know-how and business network in several fields.**

Company Mentors:Thanks to winning access to prestigious acceleration programs, such as Alimenta 2 Talent, Unicredit Start Lab and Startup Initiative, we now can rely on experienced business mentors, such as:

- **Alessandra Tedesco. Head of the trademark department at BARZANÒ & ZANARDO, Alessandra coordinates the activities of trademarks and patents attorneys. She is a member of the International Trademark Association ("INTA") and the European Community Trademark Association.**
- **Paolo Carassini - He is an Automotive and Logistics executive with particular expertise in change management and project management. He is currently Director General at Toyota Material Handling Italy, the "Italian Sales & Service Company" of the Toyota Group (forklifts and automation logistics systems).**

3.2 Work packages, deliverables, milestones, risks

Overall Structure of the Work Plan

The workplan follows a market oriented approach, in which each milestone defines a must-have feature of the product or will contribute to preparing its market uptake. It can be divided into three phases: 1) Definition of the products portfolio; 2) Upscaling of MOGU production capacity; and, 3) Preparation for market uptake.

In WP1 we will work closely with stakeholders, end users and customers in order to define the design and certifications that the first collection of MOGU floor must have in order to satisfy the market need. On the basis of the specifications obtained, in WP2 the manufacturing process of the final products will be rapidly optimized and the protocols to be used during the production established. During WP3, the upscaled production plant will be designed, built and commissioned. In parallel to this, in WP4 MOGU floor will achieve the CE mark and additional green certifications to be market ready and increase its value. Cross cutting the product development, we will continually build on and upgrade our business development plan in WP5. As part of this activity the marketing plan will review the business opportunities of the technology and devise a sales strategy (building on the initial strategy contained in the business plan). Finally, a project management framework will be implemented through WP6. The following Gantt shows the timings and the interdependencies between the different WPs and tasks (and the Deliverables-D- and Milestones-M- are also deptecited. Our team is ready to deploy all the activities needed at technical, commercial and management level thanks to our facility of 350 m^2 which is equipped with a pilot plant, a microbiology lab, two offices and an exposition space. The site is also ready to be expanded up to 2000 m^2 to host the upscaled plant and the increased staff. To prepare the market uptake of MOGU floor we are already negotiating pre-commercial

agreements with important stakeholder of the building and design Associated**sectors**with**such**document**forexample:**Ref Ares(2018)3790472**ARUP,Incatro** - 17/07/2018 **and ExT. These companies have provided Letters of Support, which have been Annexed to the end of this bid.**

Timing of the work packages and their components (include a Gantt chart or similar).

ACTIVITY	MONTHS																							
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
WP1 Designing of MOGU floor collection				M																				
Selection of product certification				D																				
Definition of the technical specifications				D																				
Definition of the aesthetic design																								
WP2 Product Manufacturing Optimization						M																		
Optimization of MOGU growing phase																								
Optimization of post-production						D																		
WP3 Production upscale and digitalisation										M								M						M
Upscaled plant design										D														
Building and commissioning of the new plant																		D						
Increase process automation and digitalization																						D		
Demonstration with early adopters																								
WP4 Certifications													M					M						M
Achievement of CE mark																								
Achievement of additional green certifications																		D						
Achievement of process certifications																								D
WP5 Gear up for commercialisation																		M						M
echnology Watch and Knowledge Management																								
Market and Business Planning									D															D
Preparation for commercialisation																		D						
Communication, dissemination and marketing			D			D																		
WP6 Management																								
Monitoring and reporting	D											D												D
Risk Management																								
Innovation Management								D								D								D

3.3 Resources

What are the resources, equipment and facilities required for the project and how will you access them?

Would you need additional resources for scaling-up your business?

Regarding the resources, equipment and facilities required for the project, these have been carefully planned in terms of the personnel that need to be mobilized internally, along with the other direct costs, travel, equipment and goods and services. Regarding the personnel resources necessary in relation to the process monitoring and control system in WP3, we will subcontract this work out to a specialized engineering company (as is described in Section 4.2).

Regarding the additional resources to scale our business, as was discussed in Section 2.3, subsequent financing in the post-project phase will be through a mix of self-financing, whereby we will invest profits from increased sales in growing and scaling our business, combined with private investment and potential additional grants in the future.

Table 3.g: 'Other direct cost' items

[SEP – 210509337] [MOGU Floor] –Part B –